AFFIAT · AFRO-ASIAN FORUM FOR INNOVATION AND TECHNOLOGY · ASWAN FINALS 2026 · GAMES

Pixel Verse Studios

Game Technical Documentation

Crab Quest and Space X: Journey to Jupiter. Two 16-bit games that teach real science.


Mariam Alaa


Selim Loai


Mohamed Loai



Medhat Ali


Salah Mohamed

1. Project Overview

Field	Value
Studio	Pixel Verse Studios
Project Type	Game studio: two desktop games and a studio website
Products	**Crab Quest** (6-chapter ocean platformer) and **Space X: Journey to Jupiter** (4-level space adventure)
Description	Pixel Verse Studios is a small indie studio that makes 16-bit narrative games for children. In each game the player learns real science without a lecture: ocean life and ocean conservation in Crab Quest, the solar system in Space X.
Target Age Group	5-99 years old
Team	Mariam Alaa, Selim Loai, Mohamed Loai, Medhat Ali, Salah Mohamed
Competition	AFFIAT · Afro-Asian Forum for Innovation and Technology · Aswan Finals 2026, Games category

The two games

	Crab Quest	Space X: Journey to Jupiter
Genre	Platformer, survival and collection in one story	Walk-and-collect, dodge and shoot
Story	A crab escapes a kitchen and journeys home through 6 chapters	A Hero flies from Earth to Jupiter's moon Europa to find out if its hidden ocean could hold life
Structure	6 chapters picked from a level-select screen	4 levels played in order, then a win screen
Window	1200 × 600	800 × 600
What you learn	Marine life, pollution and predators, coral sanctuaries	Rockets, space junk, the asteroid belt, Europa's ocean, Jupiter, and a fact about the Sun or a planet after every loss

Tech Stack

Category	Technology	Purpose
Language	Python 3.11+	All game logic
Game engine	Pygame Zero (`pgzero` 1.2.1) on Pygame 2.6.1	Window, `Actor` sprites, keyboard, mouse, music
Website	Plain HTML + one hand-written CSS file + a small JavaScript file	The studio's public page about both games
Art	16-bit pixel art (AI-assisted, cleaned by the team)	Backgrounds, sprites, planets
Font	Jersey 10 (SIL Open Font License)	Pixel text inside Space X
Packaging	PyInstaller (Windows) and itch.io	Planned release path

Category	Technology	Purpose
Source control	One Git repository holding all three projects	Shared history and issues

2. Problem Statement

Facts in school can feel far away from play. A child who is told that Europa hides an ocean, or that plastic hurts sea turtles, hears a sentence. A child who has to *dodge* that plastic, or *fly through* Europa's ice, remembers an experience.

What problem does this project solve?

The studio turns science facts into game moments, so learning does not need a separate lesson. Every level has a reason to exist in the story, and every fact shows up exactly where it makes sense in the game.

Why does it matter?

Children already spend time on games. Games that teach honestly, with real facts and no fake features, give that time a purpose. Pixel art and short sessions keep the games light enough to run on any school computer.

How is it solved?

- **Crab Quest** carries a conservation story: the crab dodges real threats (birds, trash, predators), helps a shrimp, and ends in a protected coral sanctuary.
- **Space X** carries a science mission: each level ends on a *Fact card* with a story line and one real fact, and every game over shows a picture and a fact about a planet.

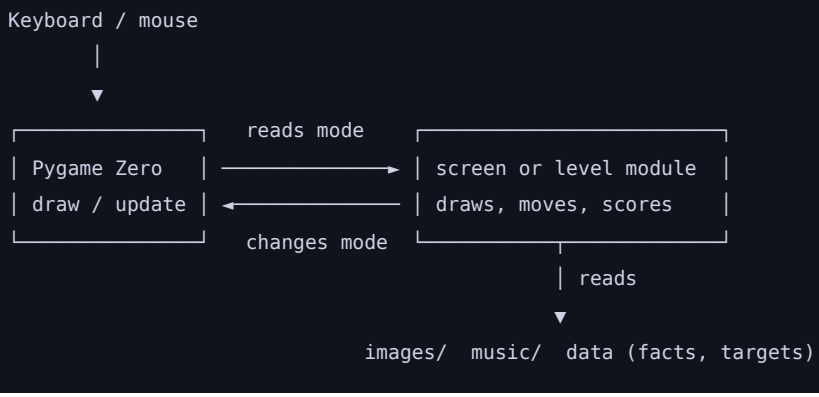
3. Technical Architecture

The studio keeps both games and the website in one repository. Each project keeps its own folder, its own `ISSUES.md` and its own instructions.

```
PVS/
├── Crab-game/              Crab Quest (Pygame Zero)
│   ├── main.py             the only entry point: hooks for draw/update/input
│   ├── game_states.py      shared state: game_mode, music, game over, win
│   ├── start_screen.py, level_select.py
│   ├── level_1.py ... level_6.py    one self-contained module per chapter
│   ├── images/, music/
├── space-game/             Space X: Journey to Jupiter (Pygame Zero)
│   ├── Game.py             only the Pygame Zero hooks, switches on the mode
│   ├── screens/            title, fact card, game over, win
│   ├── levels/             launch_pad, earth_orbit, belt, europa
│   ├── parts/              ship, controls, hud, shared game state
│   ├── data/               level targets, story lines, planet facts
│   ├── images/, music/, fonts/
├── pixel-verse-studios/    studio website (HTML + CSS)
└── docs/                   this documentation, the poster and the slides
```

Architecture and Data Flow

Both games use the same idea: a **mode switch**. One shared variable says which screen is showing, and each Pygame Zero hook (`draw`, `update`, key and mouse events) hands the work to the module that owns that screen.



Key Files

File	Purpose
`Crab-game/main.py`	Entry point. Switches on `game_states.game_mode` (`START_SCREEN`, `LEVEL_SELECT`, `LEVEL1` … `LEVEL6`) and draws the global mute button
`Crab-game/game_states.py`	Shared state. A module changes screen by assigning `game_mode`
`Crab-game/level_N.py`	One chapter: its own sprites, platforms, physics constants and `reset()`
`space-game/Game.py`	Pygame Zero hooks and `start_level()`; switches on `game_states.mode` (`TITLE`, `PLAYING`, `FACT_CARD`, `GAME_OVER`, `WIN`)
`space-game/levels/*.py`	One module per level; each decides when it is cleared
`space-game/data/level_data.py`	One number per level (points target), the story line and the fact on each Fact card

File	Purpose
`space-game/data/game_data.py`	The planet facts shown at game over: image plus lines of text
`pixel-verse-studios/index.html` etc.	The four pages of the studio website
`pixel-verse-studios/scripts/check_site.py`	Acceptance check for the website

4. Features & User Flow

Feature List

#	Feature	Game	Status
1	Six chapters with different mechanics (platforming, dodging, collecting, surviving)	Crab Quest	✅
2	Level-select screen and mute button	Crab Quest	✅
3	Walkable launch pad with a 30-second countdown and a four-item checklist	Space X	✅
4	Dodge space junk in Earth orbit	Space X	✅
5	Shoot asteroids in the belt; the Ship fires on its own	Space X	✅
6	Two-hit ice chunks on Europa, then an automatic 3-second scan	Space X	✅
7	Fact card after every level	Space X	✅
8	Planet picture and fact at every game over	Space X	✅
9	Win screen on Jupiter, high score kept during the session	Space X	✅
10	Clickable on-screen buttons next to the keyboard controls	Space X	✅
11	Windows download and itch.io page	Both	⏳

> Status: ✅ Complete | ⏳ In Progress | ❌ Not Started

Controls

Input	Crab Quest	Space X
← → (or A D)	Move	Move the Hero or the Ship
SPACE	Jump	Start, continue, and shoot in the belt and on Europa
P		Pause
R	Restart	Retry the level / fly again
ENTER	Back to level select	
F		Fullscreen

Input	Crab Quest	Space X
Mouse	Choose a level, mute	Choose on menus, on-screen left, right and pause buttons

User Flow

```
Crab Quest
  START → LEVEL SELECT → chapter 1-6 → (win or lose) → LEVEL SELECT

Space X
  TITLE → L1 Launch pad → L2 Earth orbit → L3 Asteroid belt → L4 Europa → WIN
                each level cleared → FACT CARD → next level
                all hearts lost    → GAME OVER (planet fact) → same level again
```

Screen Reference

Screen	Game	Description	Transitions To
Start	Crab Quest	Title, animated fish, start button	Level select
Level select	Crab Quest	Six chapter buttons	Any chapter
Chapters 1-6	Crab Quest	See section 5	Level select / start screen
Title	Space X	Logo, Hero, mission text, high score	Launch pad
Launch pad	Space X	Collect 4 items before the countdown ends, then board	Fact card
Earth orbit, Asteroid belt, Europa	Space X	Flying levels with hearts and a points target	Fact card or Game over
Fact card	Space X	Story line (cyan) and a science fact (white)	Next level
Game over	Space X	Random planet picture and fact	Same level again
Win	Space X	Jupiter, mission complete, fact	Title

5. Level Design

Crab Quest: six chapters

#	Chapter	Type	What to do	Win	Lose
1	Kitchen Escape	Platformer	Jump across 7 counter platforms	Reach the goal at the top right	None (retry any time)

#	Chapter	Type	What to do	Win	Lose
2	Beach Run	Survival	Dodge falling birds	Survive 30 s, then reach the ocean wave	Bird collision
3	Sea Floor Dodging	Survival	Dodge falling trash	Survive 10 s, then follow the turtle	Trash collision
4	Shrimp Help	Platformer, collection	Gather algae food as the shrimp	Collect all 17 items in 60 s	Timer runs out
5	Predator Chase	Survival	Avoid patrolling evil fish	Survive 45 s	Fish collision
6	Home Sweet Home	Platformer	Reach the crab family	Reach the family	None (final level)

Space X: four levels

#	Level	What to do	Ends when	Lose
1	Launch pad	Walk the Hero, collect helmet, ammo crate, lucky football and fuel can, then board the Ship	All 4 items collected, Hero boards, Ship blasts off	The 30-second countdown reaches 0
2	Earth orbit	Dodge space junk (the Ship cannot shoot)	10 pieces dodged	Lose all 3 hearts
3	Asteroid belt	Shoot asteroids; the Ship fires every half second	30 asteroids shot	Lose all 3 hearts
4	Europa	Break the ice (2 hits each); then a 3-second scan	12 ice chunks broken	Lose all 3 hearts

Winning level 4 flies the Hero to Jupiter and shows the win screen with the score and the session high score.

6. UI/UX Design

Both games use 16-bit pixel art with hard-edged pixels and a dark outline on every object. Light comes from the upper left. The website uses the same dark, neon "arcade" mood.

Screenshots

Space X (captured from the running game):


The Hero's mission: fly to Europa, Jupiter's icy moon, and find out if its hidden ocean could hold life.
Press Space to start


T-5
Level 1: Launch pad
Score: 0
LEFT
RIGHT
PAUSE


Score: 2
Level 2: Earth orbit
2 / 10
LEFT
RIGHT
PAUSE


Score: 7
Level 3: Asteroid belt
1 / 30
LEFT
RIGHT
PAUSE


Score: 7
Level 4: Europa
0 / 12
LEFT
RIGHT
PAUSE


YOU REACHED JUPITER!
Mission complete! The Hero scanned Europa's ocean and reached Jupiter.
Jupiter is the biggest planet. It is a gas giant with a famous Great Red Spot, a storm that has raged for hundreds of years.
Score: 40
High Score: 40
Press R to fly again

Crab Quest:









Design System

Element	Style
Page background	#0F131C (website), deep navy and black space (Space X)
Accent colors	Purple #D0BCFF, cyan #4CD7F6, green #4EDEA3
Fonts	Space Grotesk (headings) and Inter (body) on the website; Jersey 10 pixel font in Space X
Buttons	Square corners with a hard offset shadow; pixel arrows in Space X
Layout	Full-width sections with cards on the website; one 800 × 600 or 1200 × 600 window per game

Responsive Design

Breakpoint	Layout	Tested?
Games: fixed windows (800 × 600, 1200 × 600)	Desktop only; Space X also supports fullscreen with F	✅
Website: desktop	Full-width sections, card grids	✅
Website: tablet and mobile	Cards stack in one column	☐

7. Asset Inventory

Space X

Category	Files	Folder
Backgrounds	6: launch pad, Earth, belt, Europa, Jupiter, title	`images/backgrounds/`
Characters	Hero, small Hero (2 directions), Ship, Ship on the pad	`images/characters/`
Sprites	asteroid, space junk, ice, cracked ice, football	`images/sprites/`
Checklist items	helmet, ammo crate, football, fuel can	`images/items/`
Interface	left, right, pause, continue, heart, logo	`images/ui/`
Planet pictures	the Sun and 8 planets, plus the Moon and an asteroid	`images/planets/`
Audio	background music, crash sound, shoot sound	`music/`
Font	Jersey 10 (SIL Open Font License)	`fonts/`

Crab Quest

Category	Count
Backgrounds	7
Characters (crab, shrimp)	3
UI elements	5
Fish and enemies	9
Environment	5
Hazards	5
Collectibles and platforms	3
Music tracks	1
Total	**38**

8. Data Sources

All content is stored inside the projects. There are no online services, accounts or databases.

Source	Type	Description
`space-game/data/game_data.py`	Python list	Nine entries (the Sun and the eight planets): an image path and the lines of text for each fact

Source	Type	Description
`space-game/data/level_data.py`	Python list	For each of the four levels: name, background, points target, story lines and fact
`images/`, `music/`, `fonts/`	Local files	Sprites, backgrounds, planet pictures, music, sound effects, pixel font
`game_states` modules	In-memory state	Current screen, health, score; the Space X high score resets when the game closes

External APIs

API	Purpose	Rate Limit
None	Everything runs offline	N/A

9. Educational Content

Learning Objectives

By playing, a student will:

1. Learn that ocean animals face real threats: predators, trash and lost homes (Crab Quest)
2. See why protected areas such as coral sanctuaries matter (Crab Quest chapter 6)
3. Learn why rockets carry their own oxygen, and that Earth's orbit is crowded with old satellites (Space X)
4. Learn that the asteroid belt lies between Mars and Jupiter and that ships can fly through it safely (Space X)
5. Learn that Europa hides a salty ocean under its ice, and that Jupiter is a gas giant with a Great Red Spot (Space X)
6. Meet the Sun and all eight planets through the game-over facts (Space X)

Cultural / Historical Context

Topic	Description
Marine life	Crabs, shrimp, sea turtles, fish and coral reefs, and how they live together
Ocean conservation	Pollution in chapter 3, predators in chapter 5, a sanctuary in chapter 6
Space exploration	Launches, orbit, the asteroid belt and the search for life on Europa
The solar system	Facts about the Sun and each planet

Curriculum Alignment

Skill Area	What the Student Practices
Science	Ocean ecosystems, the solar system, conservation
Problem Solving	Timing jumps, dodging, planning a route through a level

Skill Area	What the Student Practices
Computational Thinking	State machines, spawn timers, collision rules, difficulty targets
Creativity	Pixel art, level layout, storytelling
Collaboration	A five-person team sharing one repository

10. Marketing Plan

> **Short version** — for full marketing plans, use the main Afro-Asian template.

Target Audience

- **Primary**: Children and families who want a short, friendly game that teaches something real
- **Secondary**: Teachers looking for science-integrated games for the classroom or the computer lab
- **Tertiary**: Indie-game fans and AFFIAT visitors

Platforms

- Windows desktop, from a PyInstaller build (in progress)
- itch.io studio page with both games and a source download (planned)
- The studio website, which describes both games

Promotion Ideas

1. Present at the AFFIAT Aswan Finals 2026 with the poster and a live demo
2. Publish both games on itch.io with screenshots, controls and downloads
3. Run a classroom session where students play and then answer "what did you learn?"
4. Share gameplay screenshots and short clips from the real games

11. Team Worksheet

Our Studio Information

Studio: Pixel Verse Studios

Games: Crab Quest, Space X: Journey to Jupiter

Date: 2026

The team

Member	Portrait
Mariam Alaa	
Selim Loai	
Mohamed Loai	
Medhat Ali	
Salah Mohamed	

What are your games about?

Crab Quest follows a crab from a kitchen counter, over a beach and the sea floor, past a predator, to a coral sanctuary where its family waits. Space X sends a Hero from a launch pad on Earth through orbit and the asteroid belt to Europa and Jupiter.

What makes the studio special?

Every fact in the games is real, and every fact appears at the moment it matters in the story. No fake features, no invented statistics.

What We Learned

1. The hardest part of making our games was: _____
2. The most fun part of making our games was: _____
3. If we had more time, we would add: _____
4. One thing we would do differently: _____

Screenshot of Our Project

See the screenshots in section 6.

Instructor Notes

Pixel Verse Studios shows that a small team can ship two complete educational games, with real facts and one clean architecture, using free tools.